

A PROJECT REPORT ON

“Multimodal Fusion of Behavioral Signals for Personality Trait Prediction”

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE
DEGREE OF

BACHELOR OF ENGINEERING (COMPUTER ENGINEERING)
(Academic Year: 2025–26)

Submitted By

Ms. Nikita Aher
Ms. Soniya Pawar
Ms. Pranjali Rasal
Ms. Sakshi Wagh

Under the Guidance of

Dr. Ranjana Dahake

DEPARTMENT OF COMPUTER ENGINEERING



MET's Institute of Engineering,
Adgaon, Nashik – 422003
SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

June, 2026

Certificate

This is to Certify that the project report entitles

“Multimodal Fusion of Behavioral Signals for Personality Trait Prediction”

Ms. Nikita Aher	Exam Seat No:
Ms. Soniya Pawar	Exam Seat No:
Ms. Pranjali Rasal	Exam Seat No:
Ms. Sakshi Wagh	Exam Seat No:

are bonafide students of this institute and the work has been carried out by them under the guidance of Dr. P. M. Yawalkar and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University for the award of the degree of Bachelor of Engineering (Computer Engineering).

Project Guide
(Dr. Ranjana Dahake)

External Examiner

HOD
(Dr. P. M. Yawalkar)

Principal
(Dr. V. P. Wani)

Date: / /

Acknowledgements

We have taken efforts in this project. However, it would not have been possible without the kind support and help of many individual and organizations. We would like to extend our sincere thanks to all of them. It gives us proud privilege to complete the project on **“Multimodal Fusion of Behavioral Signals for Personality Trait Prediction”**. We are highly indeted to our internal guide **Dr. Ranjana Dahake** for her guidance and constant supervision as well as for providing necessary information regarding the project and also for her support in completing the project.

We are also extremely grateful to our respected H.O.D. (Computer Department) **Dr. P. M. Yawalkar** and **Dr. Vaibhav Dabhade, Mr. Mahesh Korade** (Project Co-ordinator) for providing all facilities and every help for smooth progress of project work.

Ms.Nikita Aher
Ms.Soniya Pawar
Ms.Pranjali Rasal
Ms.Sakshi Wagh

Abstract

The Personality Prediction System is an intelligent application that analyzes a user's facial expressions and voice characteristics to predict their personality traits based on the OCEAN (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) model. The system allows users to log in, give consent, and record facial, Typing and voice data, which it preprocesses to remove noise and normalize for consistency. Machine learning algorithms such as Support Vector Machines (SVM) extract and refine multimodal features from both voice and facial inputs. These features are then used to generate a personality profile that reflects the user's behavioral tendencies. The system also includes an administrator module that manages data storage, model training, and performance monitoring. This project bridges psychology and artificial intelligence by providing an automated, data-driven method for personality assessment that is useful in recruitment, mental health evaluation, and user profiling applications.

Contents

1	Introduction	1
2	Literature Survey	2
3	Problem Statment	4
3.1	Objectives and Scope	4
3.2	Summary	4
4	Analysis	5
4.1	Project Plan	5
4.1.1	Project Plan for Semester I	5
4.1.2	Project Plan for Semester II	6
4.2	Requirement Analysis	7
4.2.1	Necessary Functions	7
4.2.2	Desirable Functions	7
4.3	Summary	7
5	Design	8
5.1	Software Requirement Specifications	8
5.1.1	Project Scope	8
5.1.2	Operating Environment	9
5.1.3	User Classes and Characteristics	9
5.1.4	Design and Implementation Constraints	9
5.1.5	Assumptions and Dependencies	10
5.2	System Architecture	11
5.3	External Interface Requirement	13
5.3.1	User Interfaces	13
5.3.2	Hardware Interfaces	13
5.3.3	Communication Interfaces	13
5.4	Software System Attributes	13
5.5	Nonfunctional Requirement	14
5.6	Risk Assessment	14
5.7	Data Flow Diagram	15
5.8	Summary	16
6	Modeling	17
6.1	Use Case Diagram	17
6.2	Class Diagram	19
6.3	Activity Diagram	20
6.4	Sequence Diagram	21

6.5	Summary	21
7	Implementation and Results	22
7.1	Implementation Details	22
7.2	Results	23
7.3	Summary	26
8	Testing	27
8.1	Formal Technical Review	27
8.2	Test Plan	27
8.3	Summary	28
9	Technical Specifications	29
9.1	Advantages	29
9.2	Limitations	29
9.3	Applications	30
9.3.1	Hardware Requirements	30
9.3.2	Software Requirements	30
9.4	Summary	31
10	Future Scope	32
11	Conclusion	33
	Appendix	33
A	Achievements	34
B	Plagiarism Report	35
C	Team Information	37
	Bibliography	37

List of Figures

5.1	System Architecture	11
5.2	Data Flow Diagram: Level 0	15
5.3	Data Flow Diagram: Level 1	15
5.4	Data Flow Diagram: Level 2	16
6.1	Use Case Diagram	18
6.2	class Diagram	19
6.3	Activity Diagram	20
6.4	Sequence Diagram	21
7.1	User Input Interface of the system.	24
7.2	Personality Predictor interface with drag-and-drop video upload functionality.	24
7.3	Video analysis module displaying uploaded video preview.	24
7.4	Test case showing variation in lighting and background conditions.	25
7.5	Bar chart representation of predicted OCEAN personality traits.	25
7.6	Identification of the dominant personality trait based on maximum predicted score.	25
7.7	Late Fusion Results for Big Five Personality Traits	26
B.1	Plagiarism Report - Page 1	35
B.2	Plagiarism Report - Page 2	36

List of Tables

4.1	Planner and Progress Report for Personality Traits Prediction	6
4.2	Planner and Progress Report II for Personality Traits Prediction	7
8.1	Test Plan for Personality Prediction System	28
C.1	Team Information	37

Chapter 1

Introduction

Personality plays a crucial role in understanding human behavior, decision-making, and emotional responses. Traditionally, personality assessment has relied on self-report methods such as questionnaires, surveys, and interviews. Although widely used, these approaches often suffer from limitations such as bias, subjectivity, lack of honesty in responses, and scalability issues. As a result, there is a growing need for automated, objective, and efficient methods to analyze personality traits. With the rapid advancement of artificial intelligence and the widespread use of smartphones and digital devices, large amounts of behavioral data are continuously generated. These include typing patterns, facial expressions, and voice signals, which provide valuable insights into an individual's cognitive and emotional states. Leveraging such multimodal data enables the development of intelligent systems that can analyze human behavior more accurately and in real time.

This project, titled Multimodal Fusion of Behavioral Signals for Personality Trait Prediction, proposes an AI-based system that predicts personality traits by analyzing facial expressions, voice patterns, and potentially textual or behavioral inputs. The system utilizes techniques from computer vision and speech processing to extract meaningful features from user data. These features are then combined using multimodal fusion methods to improve prediction accuracy and robustness.

The personality traits are evaluated based on the widely accepted OCEAN model (Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism), which provides a comprehensive framework for personality analysis. By integrating multiple data sources, the system overcomes the limitations of unimodal approaches and produces more reliable and consistent results. The application provides an interactive interface where users can upload or record their video and audio data. The system processes this input, performs feature extraction, and generates a detailed personality report in the form of scores and visual representations. This enables users to better understand their personality traits without the need for lengthy questionnaires. Additionally, the system includes an administrative module responsible for managing user data, monitoring system performance, and updating machine learning models. Proper data handling and security mechanisms are implemented to ensure user privacy and confidentiality.

The proposed system has wide-ranging applications in areas such as recruitment, mental health analysis, personalized recommendations, and human-computer interaction. By providing a faster, more objective, and scalable solution, this project contributes toward the development of intelligent systems that can better understand and respond to human behavior. Overall, this project aims to bridge the gap between psychology and artificial intelligence by utilizing multimodal data and advanced machine learning techniques to deliver accurate and meaningful personality insights.

Chapter 2

Literature Survey

Personality trait recognition has gained increasing attention due to its potential applications in psychology, human-computer interaction, and adaptive systems. Various studies have explored different modalities, data sources, and machine learning techniques to infer individual personality characteristics.

Kovačević et al. [1] conducted a pioneering study on personality trait recognition based on smartphone typing characteristics in natural, uncontrolled environments. Their work demonstrated that keystroke dynamics, including typing speed, pressure, and inter-key latency, can effectively predict Big Five personality traits. The study emphasized real-world data collection ("in the wild") and showed that unobtrusive smartphone sensing provides meaningful behavioral cues related to personality.

Earlier, Agastya et al. [2] presented a systematic literature review on deep learning algorithms for personality trait recognition, summarizing advancements across text, image, and multimodal datasets. Their findings highlighted that convolutional and recurrent neural networks significantly improve feature extraction and representation learning compared to traditional machine learning methods. However, they also noted challenges such as limited dataset diversity and lack of real-world generalization.

Wampfler et al. [3] further extended this domain by predicting affective states from smartphone touch and sensor data. Although their primary focus was on emotion recognition, their work demonstrated that sensor-based behavioral signals can be generalized to infer personality-related patterns. This supports the growing trend of using passive smartphone data for psychological profiling.

Alam et al. [4] explored personality prediction from social network data, particularly Facebook posts and activities. Their model combined linguistic and behavioral features to estimate Big Five traits, illustrating the effectiveness of social media footprints in psychological analysis. This work paved the way for multimodal approaches combining text, behavior, and interaction data.

In contrast, Parsons et al. [5] discussed design requirements for mobile learning environments, emphasizing user-centered adaptation. While not directly addressing personality recognition, their framework highlights the relevance of individual traits in personalized mobile experiences—an essential consideration in personality-aware systems.

Weston et al. [6] demonstrated the impact of personality traits on health outcomes, finding that traits such as conscientiousness and neuroticism predict disease onset. This connection underscores the importance of accurate personality assessment for applications beyond computing, including healthcare and preventive medicine.

Recent studies have embraced multimodal deep learning to enhance prediction accuracy. Bhin and Choi [7] proposed a self-attention-based fusion model combining audio, visual, and textual modalities for personality recognition. Their model achieved superior performance

by learning cross-modal dependencies, showing the potential of transformer-based architectures. Similarly, Ahmad et al. [8] introduced a hybrid deep learning technique for text-based personality classification using hierarchical representations, improving interpretability and robustness.

Ouarka et al. [9] developed a deep multimodal fusion framework integrating facial, vocal, and textual data to predict personality traits. Their results confirmed that combining complementary modalities yields better generalization than unimodal models. Alam and Riccardi [10] also explored multimodal personality prediction, integrating linguistic and acoustic features, and concluded that multimodal data significantly enhance trait inference reliability.

Overall, existing research indicates a transition from text-based and questionnaire-driven personality analysis toward multimodal, sensor-based, and deep learning-driven approaches. While promising, challenges remain in data privacy, ethical deployment, and model interpretability. Future directions include developing explainable AI models and leveraging real-time behavioral data for continuous personality assessment.

Summary

Recent research has shown significant advancements in personality trait recognition using mobile and multimodal data. Smartphone typing patterns have been found effective for identifying personality traits, while deep learning techniques such as CNN and RNN have enhanced behavioral analysis accuracy. Studies using smartphone sensors demonstrated strong potential in affective state prediction, and social media activity has proven useful for personality detection. Modern multimodal fusion models now integrate audio, visual, and text data to achieve higher reliability. Hybrid deep learning approaches combining multiple data sources have further improved performance, marking a shift from text-based models to deep, context-aware, and real-world personality recognition systems.

Chapter 3

Problem Statment

To design and implement a multimodal automated system to identify and analyze typing behavior, facial expressions, and voice characteristics for predicting personality traits..

3.1 Objectives and Scope

- To analyze a user's facial expressions and voice patterns to predict their personality traits.
- To generate personality insights using the OCEAN (Big Five) model.
- To provide a user-friendly interface for users and an admin panel for system management.
- To apply the system in areas such as recruitment, counseling, and behavioral analysis.
- To develop an AI-based system that automates personality assessment and reduces human bias.

3.2 Summary

The project is an AI-powered Multimodal Personality Prediction System that predicts a user's OCEAN personality traits by analyzing facial expressions, voice characteristics, and text input. The collected data undergoes preprocessing to remove noise, normalize inputs, and clean text. Features are then extracted from each modality and fused to form a unified dataset, which is used to train an SVM model. The trained model predicts the user's personality traits, and the system generates a comprehensive personality profile report. This approach ensures accurate, reliable, and ethical personality assessment through the integration of multimodal data and machine learning techniques.

Chapter 4

Analysis

This chapter describes the project plan adopted and presents the requirement analysis for the proposed system. The project is developed using the Rapid Application Development (RAD) model to enable fast prototyping and iterative improvements. Additionally, the Model View Controller (MVC) architecture is used to separate concerns between the user interface, business logic, and data handling, ensuring scalability and maintainability.

The stakeholders involved in the requirement analysis process include the project developers and potential end users such as students, recruiters, and researchers who are interested in analyzing personality traits using behavioral signals. Their inputs helped in identifying the system requirements and defining the core functionalities of the application.

4.1 Project Plan

4.1.1 Project Plan for Semester I

The following Table 4.1 presents the detailed project plan for Semester I. It outlines the various activities to be carried out during this phase along with the responsibilities assigned to each team member for different modules of the system. The plan ensures systematic progress by dividing the work into manageable tasks and setting clear timelines for completion.

During this semester, the primary focus is on understanding the problem domain of personality prediction using multimodal behavioral signals. The team conducts extensive research on existing systems, machine learning models, and multimodal fusion techniques. This phase also includes gathering functional and non-functional requirements and identifying suitable tools and technologies such as computer vision libraries, audio processing frameworks, and machine learning models. In addition, the system architecture is designed by defining different modules such as data collection, feature extraction, model training, and result visualization. Initial prototypes and workflows are also prepared to validate the feasibility of the approach.

Regular meetings and progress reviews are conducted to ensure that all team members are aligned with the project objectives and timelines. This structured approach helps in building a strong foundation for the development phase in the next semester. This phase plays a crucial role in minimizing future risks and ensures that the project development proceeds smoothly in Semester II.

Table 4.1: Planner and Progress Report for Personality Traits Prediction

Phase	Activity	Start Date	End Date	Group Members
1	Selection of Project Topic	20-07-2025	26-07-2025	Team
1	Literature Review and Research on Personality Prediction Models	27-07-2025	05-08-2025	Team
1	Requirement Analysis	07-08-2025	15-08-2025	Nikita Aher
1	System Design and UML Diagrams	17-08-2025	30-08-2025	Soniya Pawar
1	Study of Tools (Python, Scikit-learn, Pandas, NLP Libraries)	01-09-2025	05-09-2025	Pranjali Rasal
1	Dataset Collection and Preprocessing	07-09-2025	15-09-2025	Sakshi Wagh
1	Model Development (ML/NLP Models)	16-09-2025	10-10-2025	Team
1	Testing and Evaluation	11-10-2025	20-10-2025	Team
1	SRS and Design Document Preparation	21-10-2025	27-10-2025	Team
1	Project Review I	28-10-2025	29-10-2025	Team

4.1.2 Project Plan for Semester II

The following Table 4.2 illustrates the detailed project plan for Semester II. It focuses on the development, integration, testing, and deployment phases of the project. The plan clearly defines the responsibilities assigned to each team member for different modules such as frontend development, backend services, and machine learning components, ensuring proper coordination and accountability throughout the development lifecycle. During this semester, the primary focus is on transforming the designed system into a fully functional application for personality trait prediction using multimodal behavioral signals. The team follows an iterative development approach, dividing the project into multiple milestones to ensure steady progress and timely completion.

In the initial phase, core modules such as video, audio, and text data processing are implemented. Feature extraction techniques are applied to capture facial expressions, voice characteristics, and linguistic patterns. These features are then combined using multimodal fusion techniques to improve prediction accuracy. Subsequently, machine learning models are developed, trained, and optimized for predicting personality traits. The system is integrated with a user-friendly interface that allows users to upload data and view analysis results through visual representations such as graphs and scores.

Regular Formal Technical Reviews (FTR) are conducted at each milestone to evaluate code quality, system performance, and adherence to requirements. Various levels of testing, including unit testing, integration testing, and system testing, are performed to identify and resolve errors. Finally, the complete system is deployed and prepared for demonstration. Documentation is finalized, and performance improvements are incorporated based on testing results. This phase ensures that the project is robust, efficient, and ready for final evaluation.

Table 4.2: Planner and Progress Report II for Personality Traits Prediction

Phase	Activity	Start Date	End Date	Group Members
2	Define Programming Standards	01-01-2026	05-01-2026	Team
2	Data Collection and Feature Extraction	06-01-2026	20-01-2026	Team
2	NLP Processing and Text Cleaning	21-01-2026	05-02-2026	Nikita Aher
2	Personality Traits Model Development	06-02-2026	20-02-2026	Soniya Pawar
2	Model Training and Evaluation	21-02-2026	05-03-2026	Pranjali Rasal
2	UI Development and Integration	06-03-2026	20-03-2026	Sakshi Wagh
2	Testing and Bug Fixing	21-03-2026	05-04-2026	Team
2	Final Review and Documentation	06-04-2026	14-04-2026	Team

4.2 Requirement Analysis

4.2.1 Necessary Functions

The system must provide functionality to capture or upload multimodal input data such as video, audio, and textual information from users. This data serves as the primary source for analyzing behavioral patterns. Once the data is collected, the system should perform preprocessing and feature extraction to obtain meaningful information such as facial expressions from video, voice characteristics from audio, and linguistic patterns from text.

The extracted features from different modalities must be combined using appropriate multimodal fusion techniques to enhance the accuracy and reliability of predictions.

4.2.2 Desirable Functions

The system may include advanced features to improve usability and overall user experience. It can provide real-time personality analysis, allowing users to receive instant feedback based on their input data. An interactive dashboard can be incorporated to visually represent personality traits, making the results easier to interpret. Furthermore, the system may allow users to download detailed personality reports in PDF format for future reference.

4.3 Summary

In this chapter, we discussed the project planning and requirement analysis of the proposed system. The project plans for Semester I and Semester II provide a structured approach for the design, development, and implementation phases.

The necessary functions define the core operations of the system, including multimodal data collection, feature extraction, fusion, and personality prediction. These functionalities are essential to achieve the primary objective of the project.

Chapter 5

Design

This chapter describes the Software Requirement Specification (SRS) for the proposed system Machine Learning Based Personality Traits Prediction using Typing, Facial, and Voice Modalities. It explains the system architecture, external interface requirements, risk assessment strategy, and Data Flow Diagram which illustrates the working of the system.

5.1 Software Requirement Specifications

The Software Requirement Specification defines the scope of the project, operating environment, user characteristics, design constraints, and overall system functionality.

5.1.1 Project Scope

Personality prediction plays a significant role in various domains such as recruitment, mental health analysis, human-computer interaction, and behavioral analytics. Traditional personality assessment methods rely heavily on questionnaires and self-reporting techniques, which may introduce bias, lack accuracy, and are often time-consuming. The proposed system aims to overcome these limitations by predicting personality traits using multiple modalities such as typing behavior, facial expressions, and voice signals. By combining these diverse sources of behavioral data, the system enhances prediction accuracy and provides a more comprehensive and objective understanding of an individual's personality.

The system utilizes machine learning and deep learning techniques to extract meaningful features from different modalities, including typing patterns (keystroke dynamics), facial expressions (image/video analysis), and voice signals (speech features). These features are further processed and fused to generate reliable personality predictions.

The scope of the system includes:

Developing a multimodal personality prediction model: The system focuses on building a robust machine learning model that can analyze data from multiple modalities and accurately predict personality traits based on learned patterns.

Integrating different input modalities into a unified system: It ensures seamless integration of typing, facial, and voice data into a single framework, enabling efficient processing and analysis of multimodal inputs.

Providing real-time personality trait prediction: The system is designed

to process user input and generate personality insights instantly or within minimal time, enhancing user experience and usability.

Creating a user-friendly interface for interaction: A simple and interactive interface is developed to allow users to easily provide input data and view results without requiring technical expertise.

Supporting applications in recruitment and psychological analysis: The system can be utilized in various real-world scenarios such as candidate screening, behavioral analysis, and mental health assessment.

5.1.2 Operating Environment

The system is designed to operate in a web-based or desktop environment with the following setup:

- **Frontend:** React.js / Web Interface
- **Backend:** Node.js / Flask (Python)
- **Machine Learning Models:** Python (TensorFlow / PyTorch / Scikit-learn)
- **Database:** MongoDB / MySQL
- **Hardware:** System with camera, microphone, and keyboard

5.1.3 User Classes and Characteristics

The system is intended for the following users:

General Users: Individuals who want to analyze their personality traits.

Recruiters: For candidate personality assessment

Researchers: For behavioral and psychological studies

5.1.4 Design and Implementation Constraints

The proposed system faces several challenges due to the complexity of processing multimodal data. Since the system relies on facial expressions, voice signals, and typing behavior, external environmental factors can significantly impact performance. For instance, variations in lighting conditions may affect the accuracy of facial recognition, while background noise can interfere with voice analysis and reduce the quality of extracted features.

Additionally, differences in typing speed, keyboard layouts, and user behavior introduce variability in keystroke dynamics, making it difficult to maintain consistency across users. The implementation of machine learning and deep learning models also requires high computational resources, which may affect system performance, especially on devices with limited processing power.

Another critical constraint is related to data privacy and ethical considerations. Since the system collects sensitive user data such as facial images and voice recordings, it must ensure secure handling, storage, and processing of information to maintain user trust and comply with ethical standards.

Therefore, the system must be carefully designed to handle these constraints while maintaining efficiency, accuracy, and reliability.

5.1.5 Assumptions and Dependencies

The system is developed based on certain assumptions regarding the availability and quality of input data and resources. It is assumed that users will provide clear and usable input through the camera and microphone, enabling accurate feature extraction and analysis. The system also assumes the availability of a stable internet connection for web-based deployment and communication between different system components.

Furthermore, it is expected that users will have access to the required hardware components such as a functioning camera, microphone, and keyboard. The system also depends on the availability of pre-trained machine learning models and relevant software libraries for processing and prediction tasks.

These assumptions and dependencies are essential for the proper functioning of the system and achieving reliable personality prediction results.

5.2 System Architecture

The system follows a layered architecture consisting of multiple components, each responsible for a specific functionality to ensure modularity and scalability. The User Interface Layer handles user interaction and allows users to provide input data and view the generated results in an intuitive manner. The Data Acquisition Layer is responsible for capturing multimodal inputs, including typing patterns, facial expressions through camera input, and voice signals via microphone.

The architecture ensures modularity, scalability, and efficient processing of multimodal data.

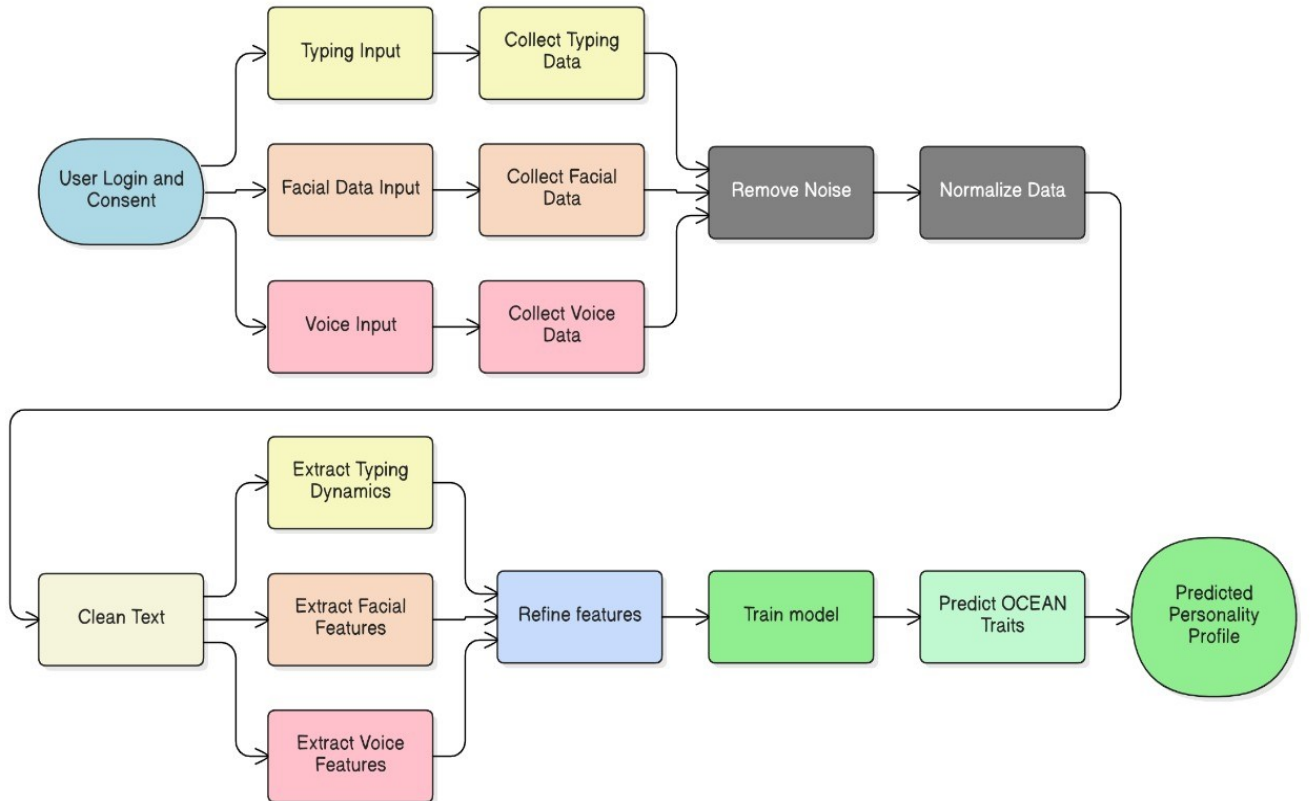


Figure 5.1: System Architecture

Multimodal Personality Traits Prediction System (OCEAN Model)

The proposed system is designed to predict human personality traits based on the Big Five (OCEAN) model using three behavioral modalities: typing behavior, facial expressions, and voice signals. The system follows an end-to-end pipeline that begins with data acquisition and ends with personality prediction and visualization using machine learning and deep learning techniques. Each modality captures a different aspect of human behavior, making the system more robust and accurate compared to single-source approaches.

1. User Login and Consent

The system starts with a secure login mechanism where the user is authenticated into the platform. After successful authentication, the system explicitly requests user consent

before collecting any behavioral data. This step is crucial as the system deals with sensitive biometric and behavioral information such as facial expressions, voice recordings, and keystroke patterns. The consent module ensures transparency, privacy protection, and ethical compliance, making the system suitable for real-world deployment.

2. Input Modalities (Data Collection Phase)

In this phase, the system collects data from three independent but parallel input streams, each representing a different behavioral dimension of the user.

Typing behavior is captured through keystroke dynamics while the user interacts with the keyboard in real time. Every key press and release event is recorded along with precise timestamps. These low-level interactions help in understanding the user's cognitive behavior, reaction time, and typing rhythm, which are often linked to personality characteristics.

Facial data is continuously captured using a webcam. The system processes video frames to extract facial expressions and micro-movements. These facial cues provide insight into emotional states and behavioral tendencies, which are important indicators of personality traits such as extraversion or neuroticism.

Voice data is collected through microphone input during user speech. The audio signal carries rich information about tone, pitch, speaking speed, and emotional expression. These vocal characteristics reflect psychological states and communication style, which are strongly correlated with personality dimensions.

3. Data Preprocessing

Before feature extraction, all collected raw data undergoes preprocessing to improve quality and consistency. For typing data, irregular timestamps and extreme delays are removed to reduce noise in behavioral patterns. Facial data is processed by filtering blurry frames, aligning faces, and normalizing lighting conditions to ensure stable feature extraction. Voice signals are cleaned using noise reduction techniques to eliminate background disturbances and silence segments. Finally, all numerical features are normalized using Min-Max scaling and Z-score standardization to ensure uniformity across different data modalities.

4. Feature Extraction

After preprocessing, meaningful behavioral features are extracted from each modality. From typing data, temporal and behavioral features such as dwell time, flight time, digraph latency, typing speed, and error correction patterns are extracted. These features represent the user's interaction rhythm with the keyboard.

From facial data, spatial and emotional features are extracted using facial landmark detection techniques. Movements of key facial regions such as eyes, eyebrows, and mouth are analyzed to determine expressions and emotional states. These features help in understanding non-verbal emotional behavior.

From voice data, acoustic and prosodic features are extracted, including MFCC coefficients, pitch variation, energy levels, speech rate, and spectral properties. These features capture vocal tone, emotional intensity, and speaking style, which are important psychological indicators.

5. Feature Fusion

Since each modality provides complementary information, the extracted features are combined into a single unified representation. This process is known as feature fusion. Early fusion is performed by concatenating all feature vectors into one combined feature space. In addition, weighted fusion is used to assign importance to each modality based on its contribution to personality prediction accuracy. To reduce redundancy and improve computational efficiency, dimensionality reduction techniques such as Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) are applied.

6. Model Training and Prediction

The fused feature vector is used to train machine learning models that map behavioral patterns to personality traits based on the Big Five (OCEAN) model. The system predicts five continuous output values corresponding to Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Each trait reflects a different dimension of personality.

Multiple machine learning algorithms are used to improve prediction accuracy. XGBoost Regressor is used as the primary model due to its strong performance on structured data and ability to handle complex non-linear relationships. Random Forest is used for robust ensemble learning, while Support Vector Machine provides effective regression in high-dimensional feature spaces. In advanced implementations, Multi-Layer Perceptron and LSTM networks are used to capture complex patterns and sequential dependencies in behavioral data. A multi-output regression framework is used to simultaneously predict all five personality traits.

7. Final Output and Visualization

The final output of the system is a quantified personality profile consisting of five numerical scores, each representing the intensity of a specific personality trait. These scores provide an interpretable psychological profile of the user based on behavioral analysis. To enhance understanding, the results are visualized using bar graphs or radar charts, allowing easy comparison of trait intensities. This visualization is implemented using frontend libraries such as Chart.js or Recharts in a web-based interface.

5.3 External Interface Requirement

5.3.1 User Interfaces

Web Interface: Allows users to input typing data, capture images/video, and record voice

Dashboard: Displays predicted personality traits and analysis results

5.3.2 Hardware Interfaces

Camera: For capturing facial expressions

Microphone: For recording voice input

Keyboard: For capturing typing patterns

5.3.3 Communication Interfaces

The system communicates over the internet using standard HTTP/HTTPS protocols to ensure secure and reliable data transmission. APIs are used to enable communication between the frontend and backend, allowing efficient data exchange and integration of machine learning services with the application interface.

5.4 Software System Attributes

Reliability: The system is designed to provide consistent and accurate personality predictions based on trained machine learning models. Proper validation and testing mechanisms

are implemented to ensure dependable outputs.

Maintainability:The system follows a modular and well-structured architecture, making it easy to update, debug, and extend with new features or improvements in the future.

Scalability:The system is capable of handling multiple users and increasing data loads efficiently without affecting performance, making it suitable for large-scale deployment.

Security:Security is a critical aspect of the system, ensuring that sensitive user data such as facial images, voice recordings, and typing behavior is securely stored and transmitted using encryption and authentication mechanisms.

5.5 Nonfunctional Requirement

Performance:The system is expected to process multimodal inputs efficiently and provide results within a reasonable time, ensuring a smooth user experience.

Accuracy:High prediction accuracy is maintained by using well-trained machine learning models and appropriate feature extraction techniques from multiple modalities.

Usability:The system provides a simple and user-friendly interface that allows users to easily interact with the application without requiring technical expertise.

Privacy:The system ensures the protection of sensitive user data by implementing secure data handling, storage, and access control mechanisms.

5.6 Risk Assessment

Data Privacy Risk:There is a high risk associated with handling sensitive user data such as facial images and voice recordings. Proper encryption and secure storage mechanisms are required to mitigate this risk.

Model Accuracy Issues:The system may produce inaccurate predictions if trained on insufficient or low-quality data. Continuous model training and validation are necessary to improve performance.

Hardware Limitations:Poor-quality cameras or microphones may affect the input data, leading to reduced system accuracy and performance.

Integration Challenges:Combining multiple modalities such as typing, facial, and voice data into a unified system can be complex and requires careful design and implementation.

Performance Bottlenecks:The use of computationally intensive machine learning models may lead to slower processing times. Optimization techniques are required to maintain efficiency.

5.7 Data Flow Diagram

The Data Flow Diagram (DFD) of the proposed Multimodal Personality Prediction System illustrates the flow of information within the system. It represents how input data is collected from users, processed through different stages, and transformed into meaningful output in the form of personality trait predictions. The DFD helps in understanding the movement of data between various system components, including input sources, processing modules, and output results.

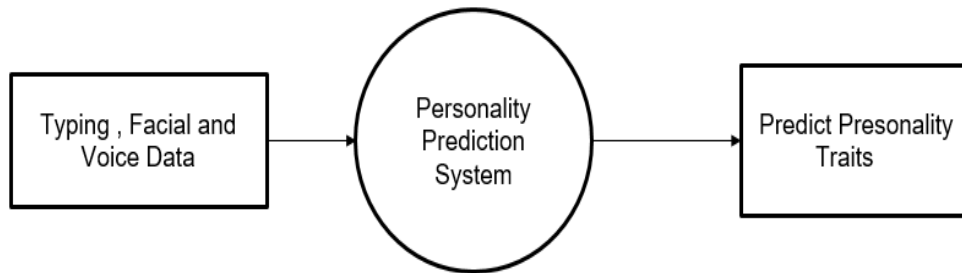


Figure 5.2: Data Flow Diagram: Level 0

Fig 5.2 provides a high-level overview of the entire system. It represents the system as a single process interacting with external entities. In this case, the user provides input data such as typing behavior, facial expressions, and voice recordings. The system processes this data and generates personality trait predictions as output. This level shows only the overall input and output flow without detailing internal processes.

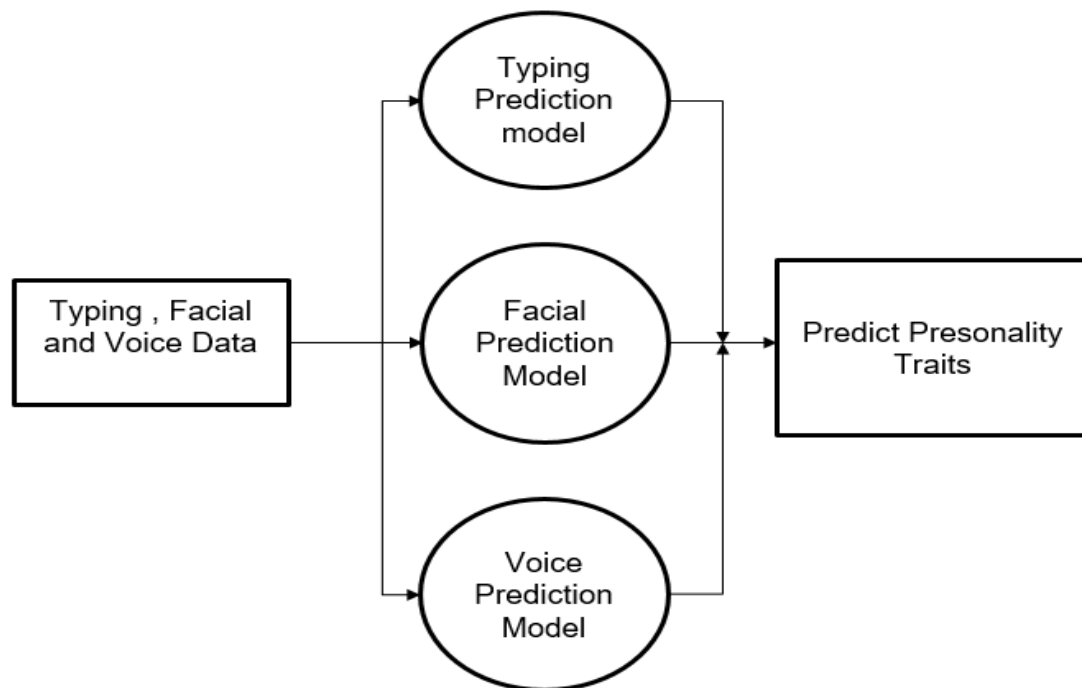


Figure 5.3: Data Flow Diagram: Level 1

Fig 5.3 provides that diagram breaks down the main system into major functional components. It illustrates processes such as data collection, preprocessing, feature extraction, model prediction, and result generation. The user input is first collected and then passed through preprocessing modules where noise removal and normalization take place. After that, feature extraction is performed for each modality, and the processed data is forwarded to the prediction module. Finally, the results are generated and displayed to the user.

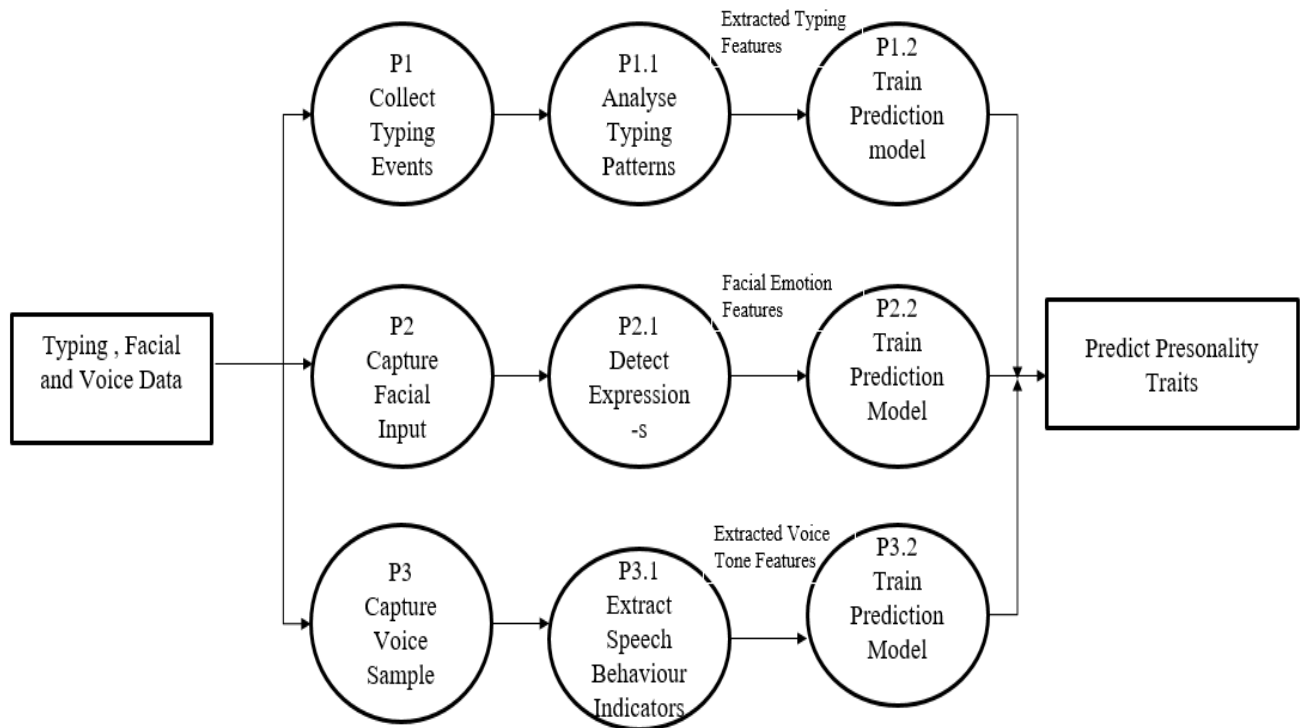


Figure 5.4: Data Flow Diagram: Level 2

Fig 5.4 provides a detailed view of the internal working of each process in the system. It includes specific steps such as keystroke analysis for typing data, face detection and expression recognition for visual data, and feature extraction from voice signals. The extracted features from all modalities are combined using multimodal fusion techniques and passed to the machine learning model for personality prediction. The final output is then stored in the database and presented to the user in a structured format.

5.8 Summary

In this chapter we studied the operating environment and the user classes and characteristics which describes the scope of the project. We have also described the software system attributes and various nonfunctional requirements.

Chapter 6

Modeling

This chapter describes the various modeling techniques used in the system **Machine Learning Based Personality Traits Prediction using Typing, Facial, and Voice Modalities**. It explains the different users of the system and the functionality of its components using UML diagrams such as Use Case, Class, Activity, and Sequence diagrams.

6.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between users, administrators, and the Multimodal Personality Prediction System. It highlights the key functions of the system including data collection, preprocessing, feature extraction, prediction, and profile viewing.

- **Actors:** System User: Provides consent, submits facial, voice, and text data, and views the predicted personality profile.
System Administrator: Manages datasets, triggers model training, and monitors system performance.
- **Use cases:** A use case defines behavioral features of a system. Login Provide Consent, Submit Facial Input, Submit Voice Input, Clean Text Data (implicit step for system processing), Predict OCEAN Traits, View Predicted Profile, Manage Data Models (Admin), Monitor System Performance (Admin)
- **Associations:** An association is a relationship between an actor and a use case. The relationship is represented by a line between an actor and a use case.
- **The include relationship:** It is analogous to a call between objects. One use case requires some type of behavior which is fully defined in another use case.

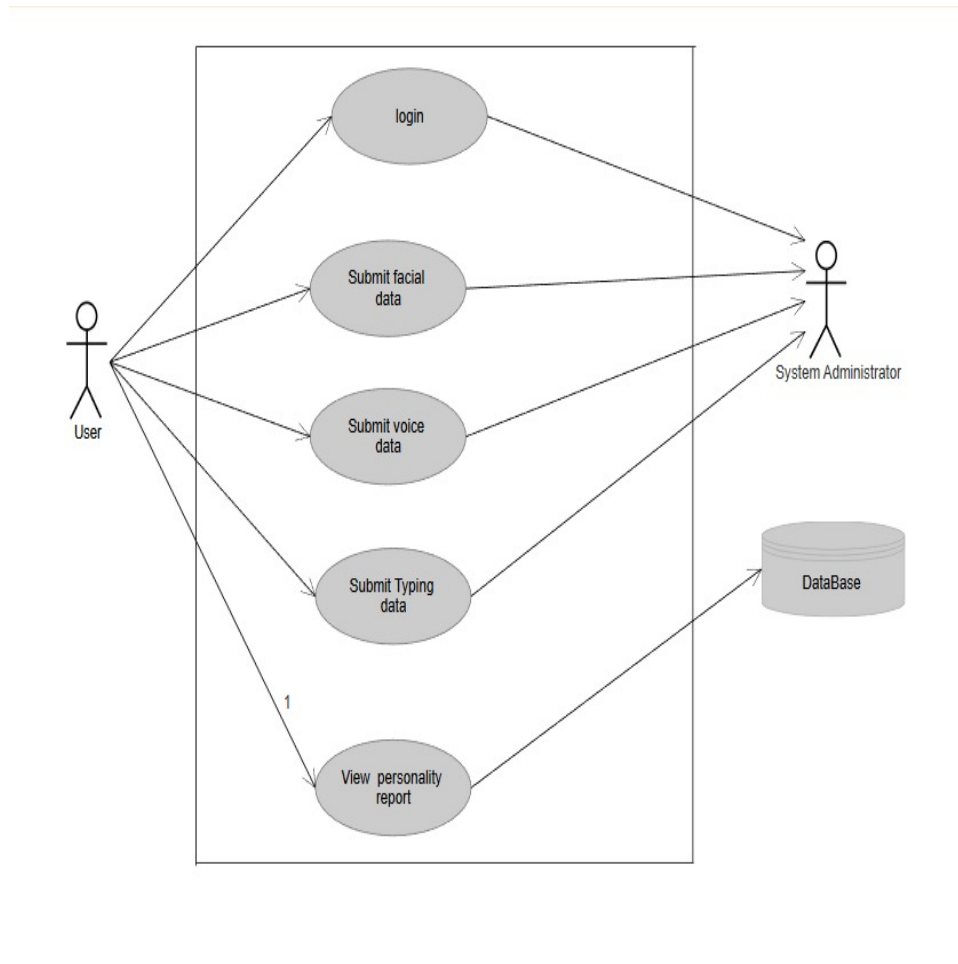


Figure 6.1: Use Case Diagram

The Use Case Diagram represents the functional requirements of the proposed Multimodal Personality Prediction System by illustrating the interaction between users and the system. It identifies the different types of users (actors) and the actions they can perform within the system. This diagram helps in understanding how the system operates from the user's perspective and defines the scope of functionalities provided.

Another important actor is the Administrator, who is responsible for managing the system. The administrator handles tasks such as maintaining user data, monitoring system performance, updating machine learning models, and ensuring the proper functioning of the application. The Use Case Diagram thus provides a clear overview of system interactions and helps in identifying the roles and responsibilities of different actors, ensuring a well-structured and user-centric system design.

Fig. 6.1 illustrates the interaction between the user and the system, where the user provides multimodal input data and receives personality predictions, while the administrator manages system operations and data.

6.2 Class Diagram

The Class Diagram presents the system structure, showing entities, attributes, methods, and relationships. It incorporates all multimodal components: facial, voice, and text data, as well as the core SVM model and personality profile. The system collects multimodal inputs through the FacialData, VoiceData, and TextData classes, which are preprocessed using the Preprocessor and TextCleaner classes to remove noise and normalize the data.

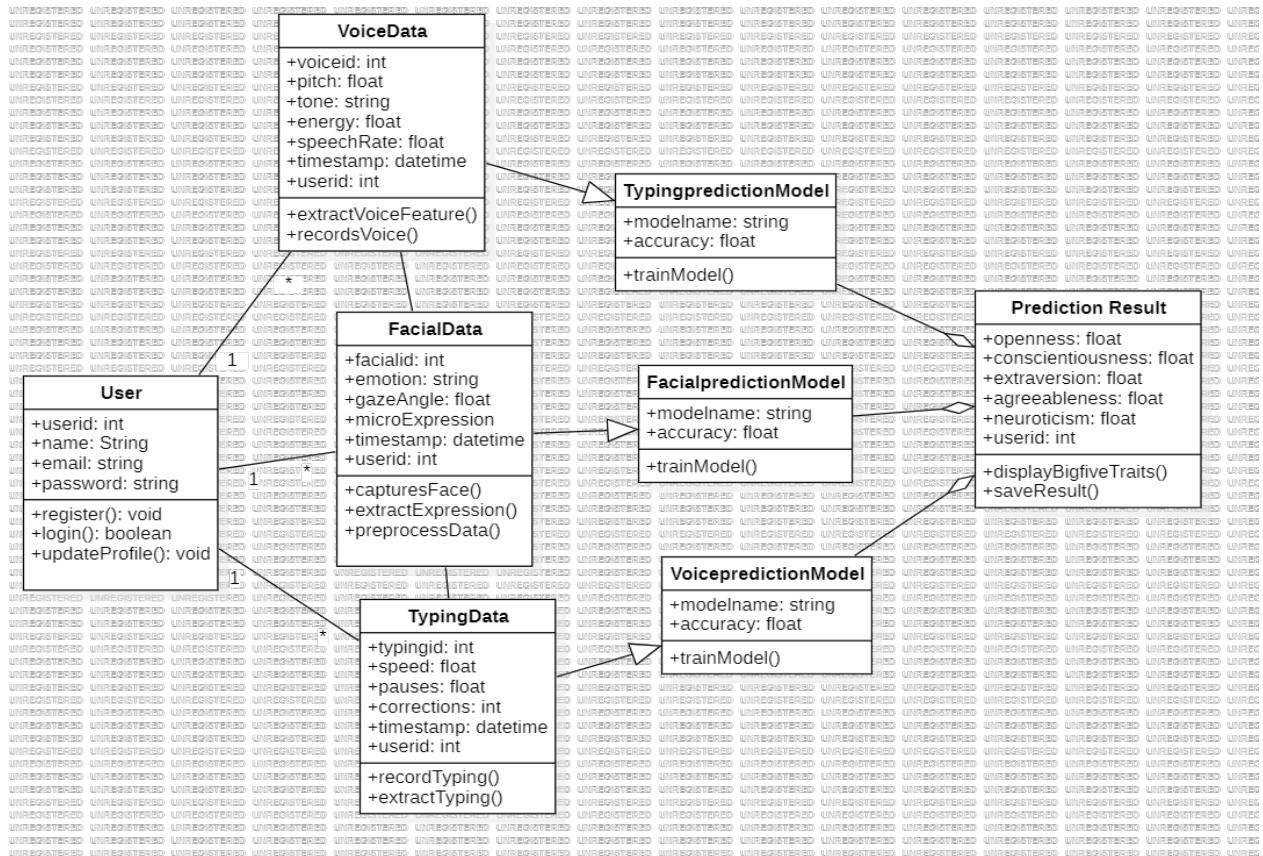


Figure 6.2: class Diagram

The Figure 6.2 illustrates The Class Diagram of the Multimodal Personality Prediction System illustrates the static structure and core components of the system, highlighting the relationships between users, data, and processing modules.

6.3 Activity Diagram

The Activity Diagram depicts the step-by-step workflow for the personality prediction process, from user login to profile generation. It clearly shows parallel processing of the three data modalities and decision points.

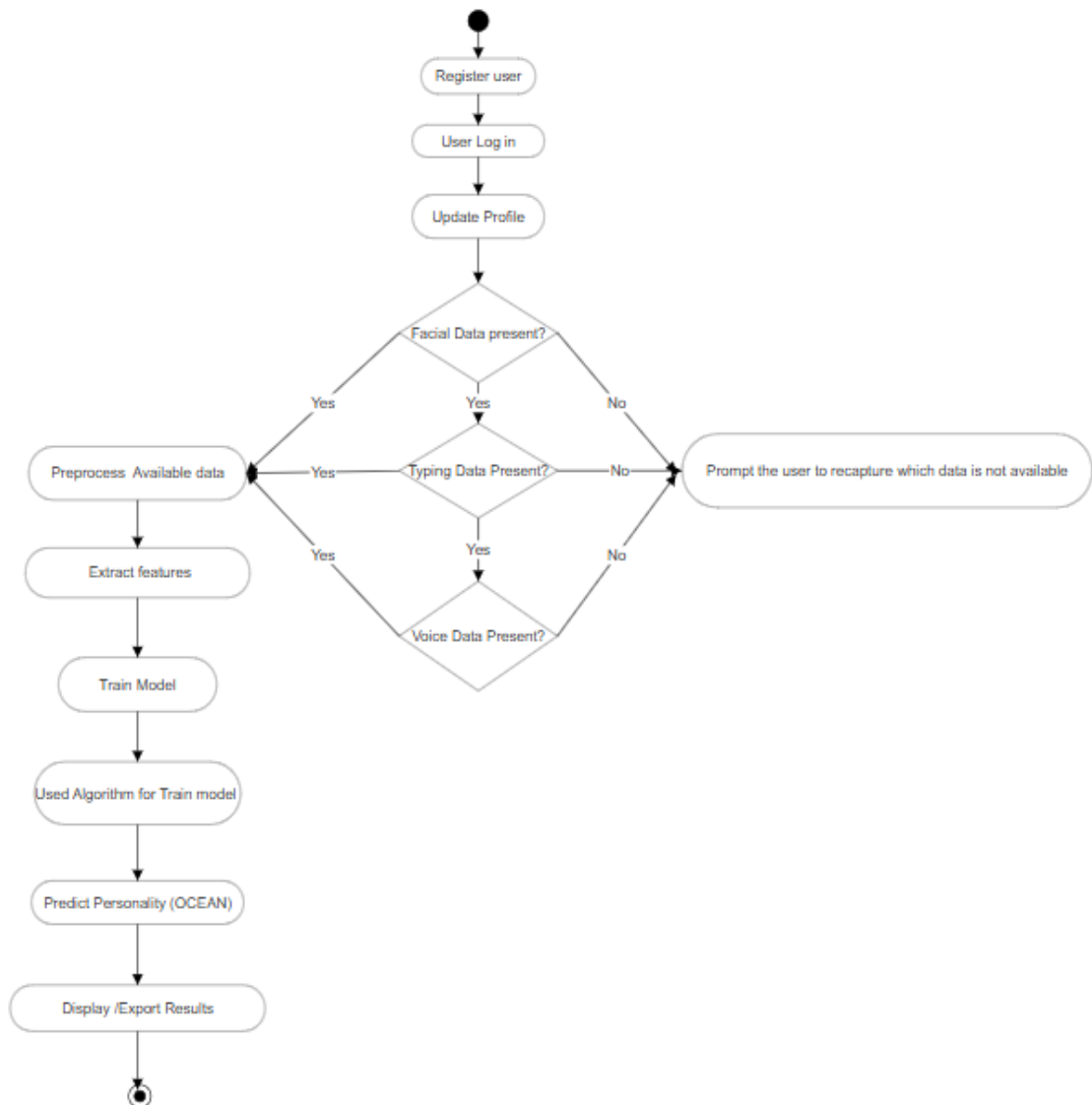


Figure 6.3: Activity Diagram

Fig. 6.3 illustrates the workflow of the system, where the user logs in, provides multimodal input data such as typing, facial, and voice signals, which are then processed through feature extraction and machine learning models to generate and display personality predictions.

6.4 Sequence Diagram

The Sequence Diagram shows the interactions between the user, system modules, and the prediction model over time. It emphasizes the flow from data submission to final personality profile generation.

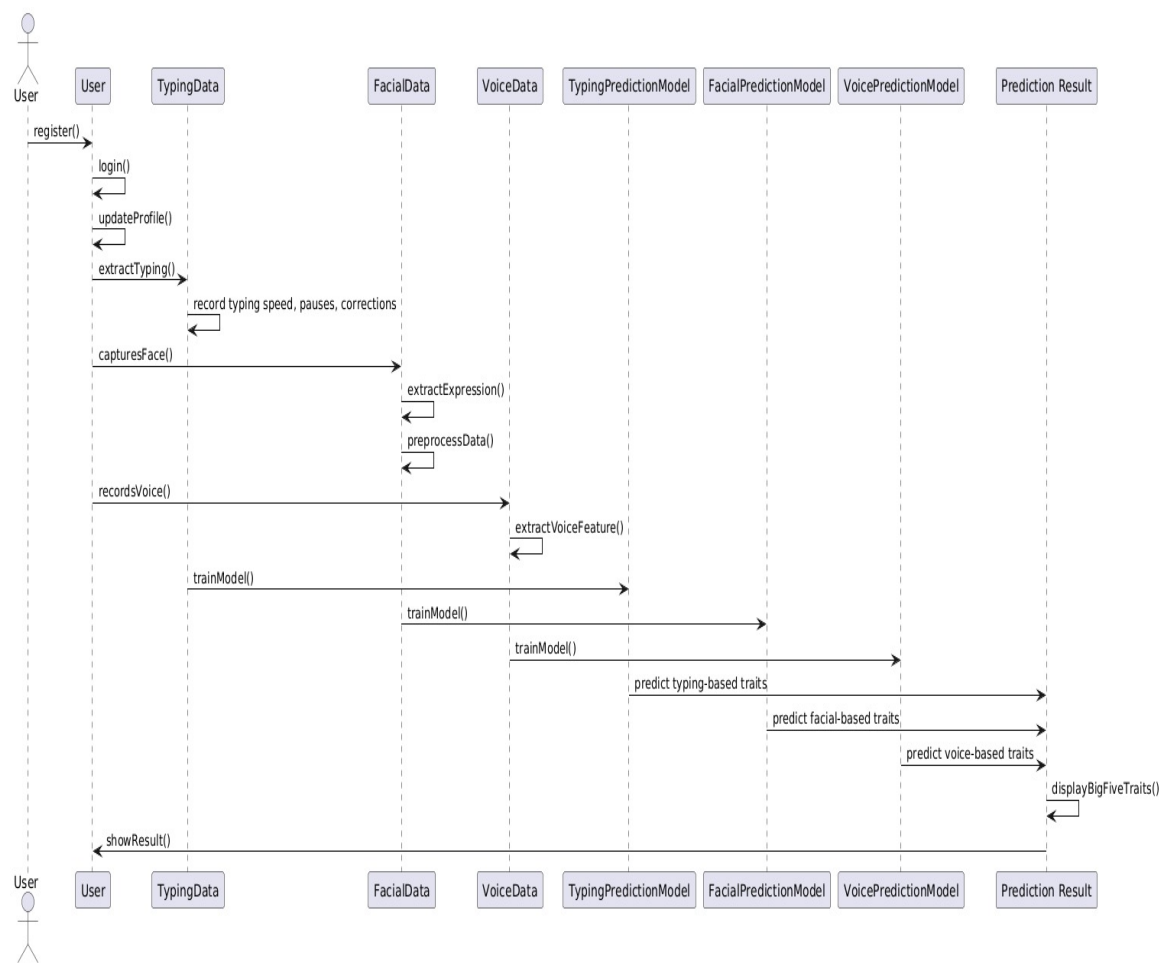


Figure 6.4: Sequence Diagram

Fig. 6.4 illustrates the sequence of interactions between the user, frontend, backend, and machine learning model, where the user sends input data, the system processes it step-by-step, and finally returns the predicted personality traits as output.

6.5 Summary

In this chapter, we explored the various modeling techniques used for the design of the proposed Multimodal Personality Prediction System. Different diagrams such as the Data Flow Diagram, Use Case Diagram, Activity Diagram, and Sequence Diagram were used to represent the system structure, workflow, and interactions.

Chapter 7

Implementation and Results

This chapter presents the implementation details and results of the system Machine Learning Based Personality Traits Prediction using Typing, Facial, and Voice Modalities. It describes the core features of the system along with their implementation techniques and outputs.

7.1 Implementation Details

This section explains the major modules and features of the system along with their implementation details:

Typing Behavior Analysis: The typing behavior module captures keystroke dynamics such as typing speed, key press duration, and latency between keystrokes. These features are collected using JavaScript event listeners in the frontend. The captured data is sent to the backend where feature extraction is performed. Machine learning models such as Logistic Regression and Random Forest are used to analyze typing patterns and map them to personality traits.

Facial Expression Analysis: The facial analysis module uses the system camera to capture user images or video streams. OpenCV and deep learning models (such as CNNs) are used for detecting facial features and expressions. Pre-trained models are used to extract emotions such as happiness, sadness, anger, and neutrality. These emotional features are further used to infer personality traits.

Voice Analysis: The voice module records audio input through the microphone. Speech features such as pitch, tone, frequency, and MFCC (Mel-Frequency Cepstral Coefficients) are extracted using Python libraries like Librosa. These features are processed using machine learning models to identify patterns related to personality traits such as extroversion or neuroticism.

Multimodal Data Fusion: The system combines data from typing, facial, and voice modalities to improve prediction accuracy. Feature-level and decision-level fusion techniques are applied. The final personality prediction is generated by aggregating outputs from all modalities using weighted averaging or ensemble learning techniques.

Machine Learning Model Implementation: Various machine learning and deep learning models are implemented using Python libraries such as Scikit-learn, TensorFlow, and PyTorch. The models are trained on datasets related to personality traits (e.g.,

Big Five traits). The trained models are deployed using Flask APIs for real-time prediction.

User Interface: The frontend is developed using React.js, providing an interactive interface where users can:

- Enter typing input
- Record voice samples
- Capture facial data
- View personality prediction results

Backend and API Integration: The backend is developed using Node.js/Flask. APIs are used to send input data from frontend to backend and return prediction results. REST APIs ensure smooth communication between modules.

Database Management: The system uses MongoDB/MySQL for storing user data, extracted features, and prediction results. This helps in tracking user history and improving model performance.

Logging and Monitoring: The system logs user activities, input data, and prediction outputs. Logging helps in debugging, performance monitoring, and improving model accuracy.

graphicx float placeins

7.2 Results

The system was tested on multiple users under different conditions. The results demonstrate the effectiveness of multimodal personality prediction.

Following are the observed results:

- The system successfully captures typing, facial, and voice inputs in real-time.
- Individual models provide moderate accuracy, while multimodal fusion significantly improves overall accuracy.
- The system predicts personality traits based on the Big Five Personality Model: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism
- The average prediction accuracy achieved is approximately 75% – 85% depending on input quality.
- The system performs efficiently with minimal response time.

Snapshots of the system include:

- User input interface (typing, voice, facial capture)
- Processing and feature extraction stage
- Final personality prediction dashboard

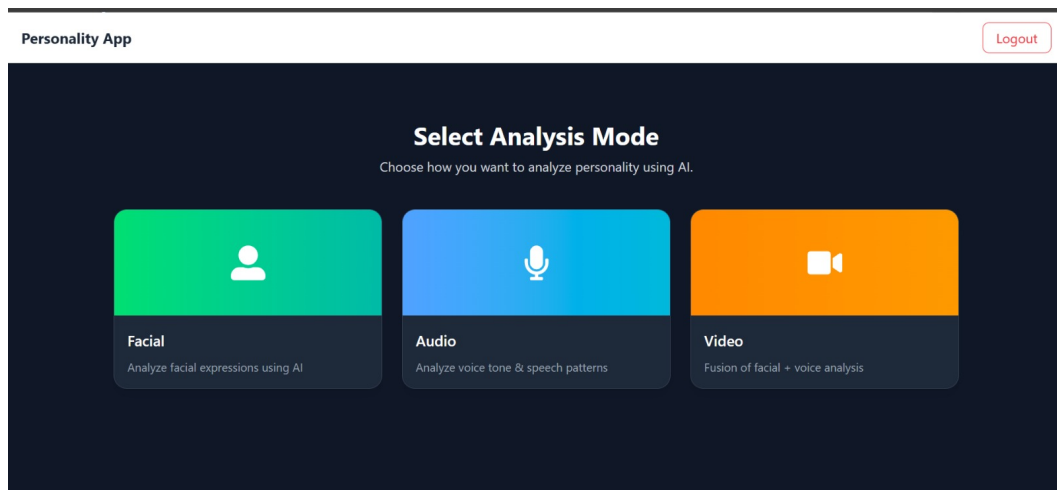


Figure 7.1: User Input Interface of the system.

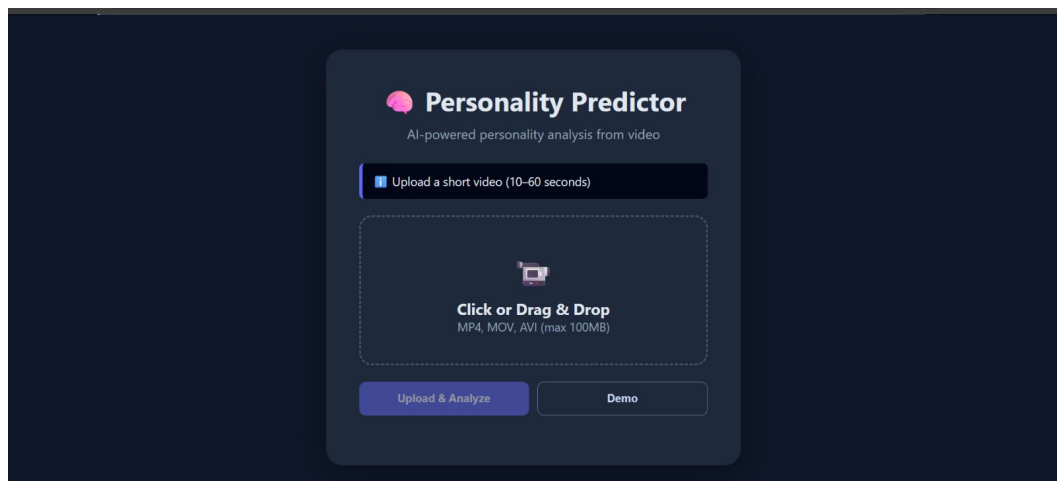


Figure 7.2: Personality Predictor interface with drag-and-drop video upload functionality.

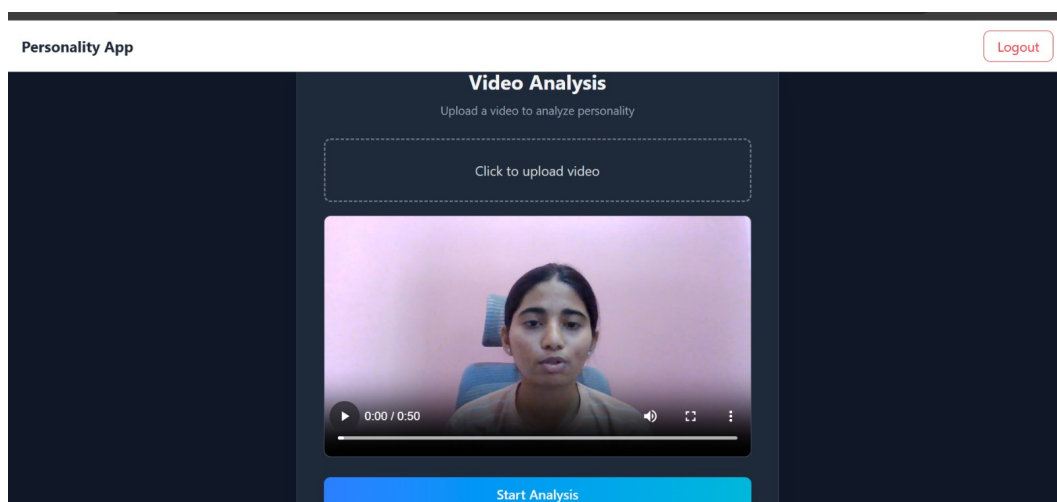


Figure 7.3: Video analysis module displaying uploaded video preview.

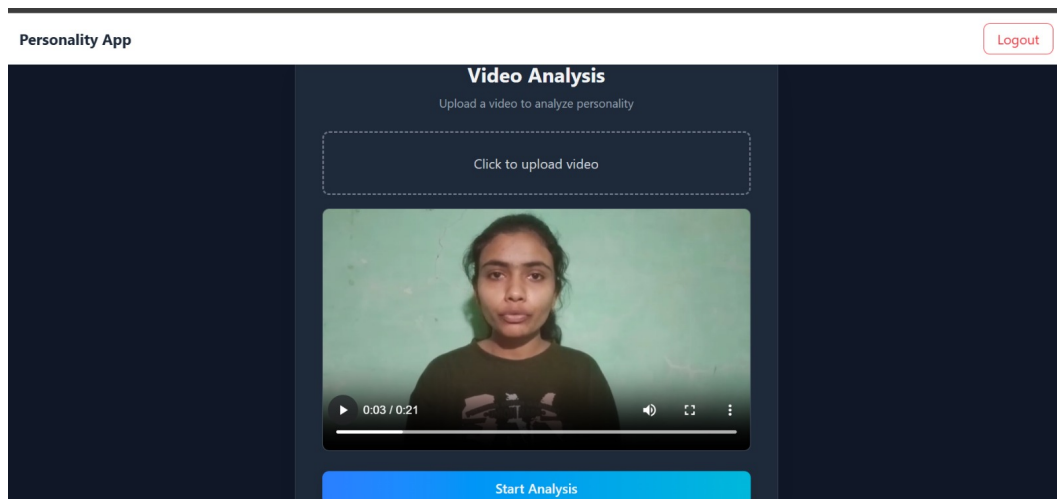


Figure 7.4: Test case showing variation in lighting and background conditions.



Figure 7.5: Bar chart representation of predicted OCEAN personality traits.

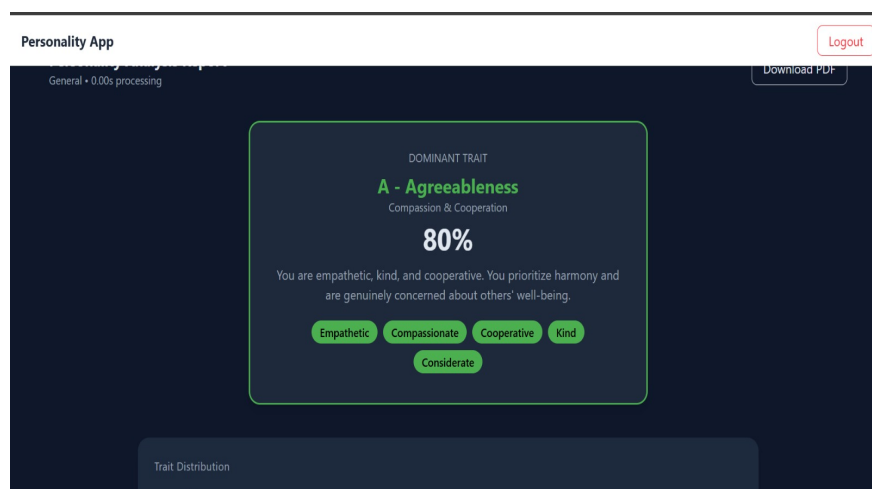


Figure 7.6: Identification of the dominant personality trait based on maximum predicted score.

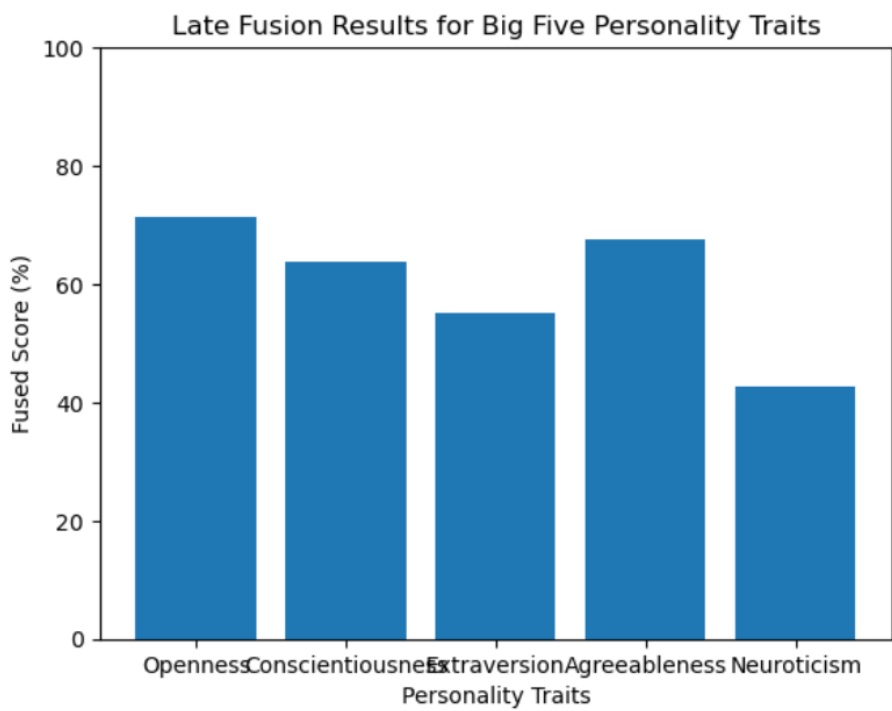


Figure 7.7: Late Fusion Results for Big Five Personality Traits

Fig. 7.7 illustrates the graphical representation of the results shown in Table 7.1. The bar graph clearly compares the performance of typing, facial, voice, and fused modalities for each personality trait, making it easier to visualize the variation in prediction accuracy among the different methods.

7.3 Summary

In this chapter, we discussed the implementation details of various modules including typing analysis, facial recognition, voice processing, and multimodal fusion. The results demonstrate that combining multiple input modalities improves prediction accuracy and system robustness. The proposed system effectively predicts personality traits using machine learning techniques while maintaining efficient performance.

Chapter 8

Testing

This chapter describes the testing strategies adopted for the system Machine Learning Based Personality Traits Prediction using Typing, Facial, and Voice Modalities. It includes details of Formal Technical Reviews (FTR) and the test plan used to validate the functionality, performance, and reliability of the system.

8.1 Formal Technical Review

Formal Technical Reviews (FTRs) are conducted to identify defects at early stages and ensure quality in both software design and implementation. Regular review meetings and discussions were carried out throughout the development lifecycle to evaluate progress and maintain consistency with system requirements. The review process involved verification of system architecture, validation of machine learning models, and evaluation of code quality. Emphasis was placed on ensuring that the system is modular, scalable, and free from logical or runtime errors. Proper coding standards, naming conventions, and documentation practices were followed to enhance readability and maintainability.

During the review process, special attention was given to the correctness of the machine learning pipeline, including data preprocessing, feature extraction, model training, and prediction accuracy. Additionally, API integration between frontend and backend components was carefully tested to ensure seamless communication and data flow within the system. The review checklist focused on key aspects such as code structure and modularity, proper use of naming conventions, adequacy of documentation, elimination of redundant or unused components, correctness of model implementation, proper API integration, and the absence of runtime errors.

8.2 Test Plan

The system was tested module-wise using pre-collected datasets consisting of images, audio recordings, and video samples. Since the system does not rely on real-time data capture, testing focused on validating the processing, feature extraction, multimodal fusion, and prediction accuracy using stored input data. Each module was verified individually to ensure correct functionality and seamless integration within the system.

Module Tested	Expected Result	Actual Result	Verdict
Image Input Module	System loads and processes facial image/video dataset.	Image/video data loaded and processed successfully.	PASS
Audio Input Module	System loads and processes pre-recorded audio files.	Audio files processed correctly.	PASS
Video Processing Module	System extracts frames and facial features from video input.	Frames extracted and processed successfully.	PASS
Feature Extraction	Extracts features from input data.	Features extracted correctly.	PASS
ML Prediction Module	Predicts personality traits.	Predictions generated successfully.	PASS
Multimodal Fusion	Combines all inputs.	Output generated correctly.	PASS
User Interface	Displays results.	UI works properly.	PASS
API Integration	Handles communication.	APIs working correctly.	PASS

Table 8.1: Test Plan for Personality Prediction System

8.3 Summary

In this chapter, we discussed the testing strategies and validation process of the system. Formal technical reviews ensured code quality, while module-wise testing verified that all system components function correctly.

Chapter 9

Technical Specifications

This chapter discusses the advantages, limitations, applications, and technical requirements of the proposed system.

9.1 Advantages

The proposed system offers several advantages that improve the efficiency and reliability of personality prediction.

Multimodal Analysis:The system combines multiple data sources such as typing behavior, facial expressions, and voice signals. This multimodal approach enhances prediction accuracy by capturing diverse behavioral patterns rather than relying on a single input source.

Real-time Prediction:The system is capable of generating personality predictions quickly after processing the input data, providing users with instant feedback and improving overall user experience.

Non-Intrusive Approach:Unlike traditional methods that depend on lengthy questionnaires, the system analyzes natural user behavior, making the process more convenient and less time-consuming.

Automation:The system automates the process of personality assessment using machine learning techniques, reducing manual effort and minimizing human bias in evaluation.

Scalability:The system is designed to be scalable and can be extended to various domains such as recruitment, psychological analysis, and human-computer interaction.

9.2 Limitations

Despite its advantages, the system has certain limitations that may affect its performance and applicability.

Data Quality Dependency:The accuracy of the system depends heavily on the quality of input data. Poor lighting conditions in images or background noise in audio can negatively impact feature extraction and prediction results.

Hardware Dependency: The system requires access to hardware components such as a camera, microphone, and keyboard. Limited or low-quality hardware may reduce system effectiveness.

Privacy Concerns: Since the system processes sensitive data such as facial images and voice recordings, it raises privacy concerns and requires secure data handling and storage mechanisms.

Model Bias: The performance of the machine learning model depends on the quality and diversity of the training dataset. Biased or insufficient data may lead to inaccurate or unfair predictions.

9.3 Applications

The proposed system can be applied in various domains where understanding human personality and behavior is important.

Recruitment and Candidate Screening: The system can assist recruiters in evaluating candidates by analyzing their personality traits through behavioral signals. This helps in making more informed hiring decisions and selecting candidates who are better suited for specific roles.

Psychological and Behavioral Analysis: The system can be used by researchers and psychologists to study human behavior and personality patterns. It provides an automated and data-driven approach for analyzing psychological traits.

Human-Computer Interaction Systems: The system enhances human-computer interaction by enabling machines to understand user behavior and respond accordingly. This can improve user experience in intelligent systems and adaptive interfaces.

Personalized Recommendation Systems: The system can be integrated into recommendation platforms to provide personalized suggestions based on user personality traits, improving the relevance and effectiveness of recommendations.

9.3.1 Hardware Requirements

- Intel/AMD Processor (i3 or above)
- Minimum 4GB RAM (8GB recommended)
- Web Camera
- Microphone
- Keyboard for typing input
- Minimum 10GB free disk space

9.3.2 Software Requirements

- Operating System: Windows/Linux/macOS

- Frontend: React.js / HTML, CSS, JavaScript
- Backend: Node.js / Flask
- Programming Language: Python
- Libraries: OpenCV, Librosa, Scikit-learn, TensorFlow/PyTorch
- Database: MongoDB / MySQL
- Tools: VS Code, Jupyter Notebook

9.4 Summary

In this chapter, we discussed the testing methodology adopted for validating the system along with its advantages, limitations, and real-world applications. The testing process ensured that all modules function correctly and meet the required performance standards.

Additionally, the advantages and limitations provided insight into the strengths and constraints of the system, while the applications highlighted its practical relevance in various domains. Overall, these specifications offer a comprehensive understanding of the system's capabilities, performance, and potential areas for improvement.

Chapter 10

Future Scope

The proposed system for Machine Learning Based Personality Traits Prediction using Typing, Facial, and Voice Modalities demonstrates promising results in multimodal personality analysis. However, there are several opportunities for further enhancement and expansion in the future. Some of the possible future improvements are listed below:

Improved Model Accuracy: Future work can focus on training the system using larger and more diverse datasets to improve the accuracy and generalization of personality prediction models.

Deep Learning Enhancements: Advanced deep learning architectures such as Transformers and multimodal neural networks can be implemented for better feature extraction and fusion.

Real-time Continuous Monitoring: The system can be extended to continuously monitor user behavior and dynamically update personality predictions in real time.

Integration with Recruitment Systems: The system can be integrated with HR platforms to assist in candidate screening and hiring decisions.

Cross-cultural Adaptation: The system can be improved to handle cultural and linguistic variations in personality traits across different regions.

Chapter 11

Conclusion

In this project, we designed and developed a system for Machine Learning Based Personality Traits Prediction using Typing, Facial, and Voice Modalities. The system leverages machine learning and deep learning techniques to analyze multimodal behavioral data and predict personality traits based on the Big Five (OCEAN) personality model. The main objective of the system is to move beyond traditional questionnaire-based assessment and provide an automated, objective, and data-driven approach for personality analysis. The proposed system integrates three behavioral modalities—typing behavior, facial expressions, and voice signals—into a unified framework. Each modality captures different aspects of human behavior, and their combination helps in building a more complete representation of personality traits. The fusion of these modalities improves the robustness of the system by reducing dependency on a single source of input and enhancing prediction reliability.

The system was implemented using Python along with machine learning and deep learning libraries, and was evaluated using image, audio, and video datasets. The processing pipeline included data preprocessing, feature extraction, multimodal fusion, and classification using trained models. The experimental results show that the proposed approach achieves improved performance compared to individual modality-based models. In terms of accuracy, the typing module improved from 70% in the base model to 75% in the proposed system. The voice module achieved 77% accuracy, while the facial module achieved 75% accuracy. The fusion-based approach, which combines multiple modalities, achieved the highest accuracy of 82%, demonstrating that combining behavioral signals leads to more stable and reliable predictions. This clearly highlights the effectiveness of multimodal fusion in enhancing system performance.

The system also provides a structured output in the form of personality trait scores based on the OCEAN model, making the results easy to interpret and useful for real-world applications. The use of machine learning ensures that the system can learn patterns from data and improve its predictive capability over time with better datasets. The project successfully demonstrates how machine learning and multimodal behavioral analysis can be used to develop intelligent systems for personality prediction. It bridges the gap between psychological assessment and artificial intelligence by providing a scalable, automated, and efficient solution. The system has strong potential for future enhancement and can be applied in domains such as recruitment, psychological evaluation, personalized recommendation systems, and human-computer interaction.

Appendix A

Achievements

- **Participation in Poster Presentation Competition**

We participated in a poster presentation competition, where we presented our project and received appreciation for our work and innovative approach.

- **Participation in National Level Project Competition**

We actively participated in a national-level project competition, which provided us with valuable exposure and an opportunity to showcase our project at a broader platform.

- **Selection for IEEE Publication**

Our project has been selected for publication under IEEE, highlighting its research contribution, innovation, and technical significance.

Appendix B

Plagiarism Report

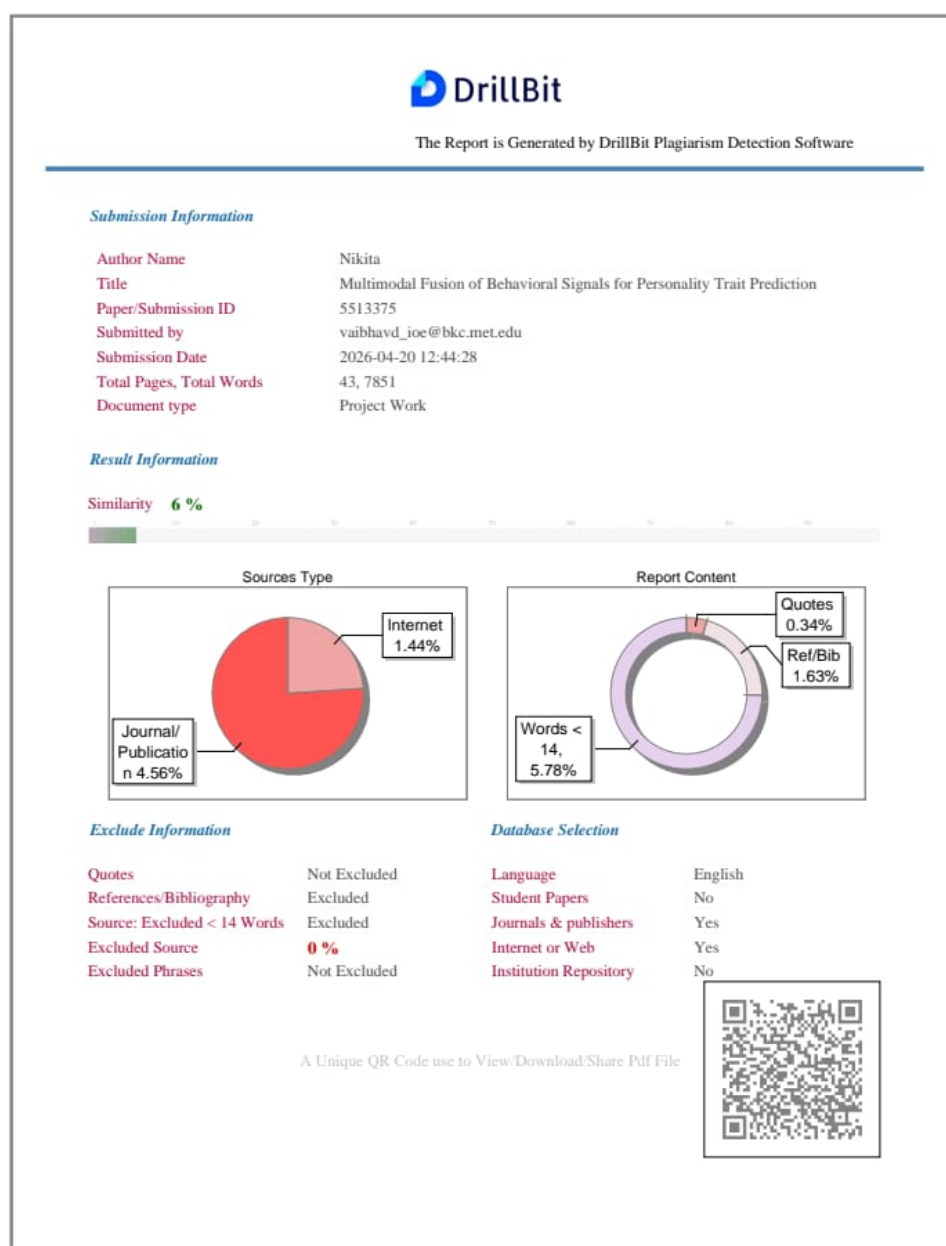


Figure B.1: Plagiarism Report - Page 1

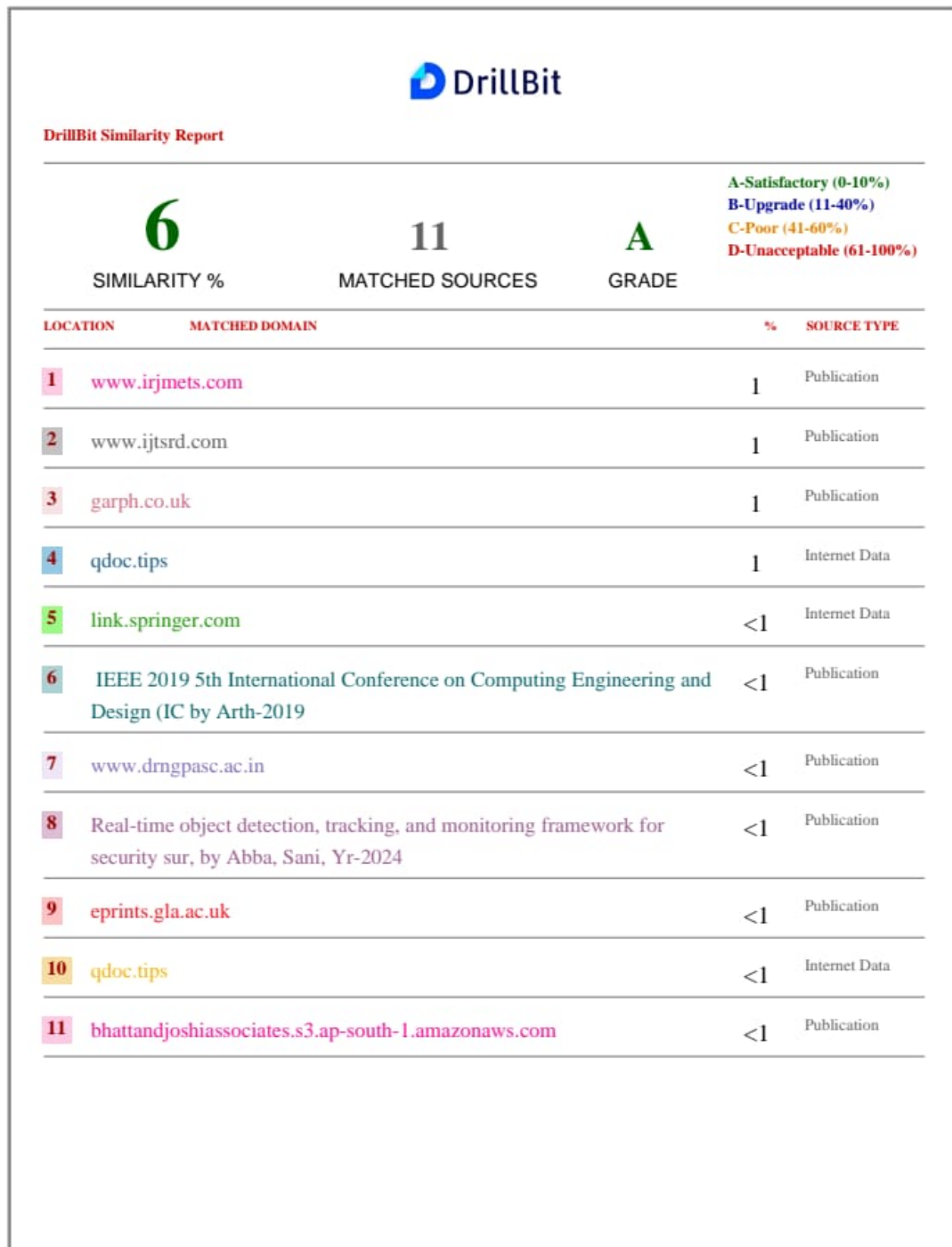


Figure B.2: Plagiarism Report - Page 2

Appendix C

Team Information

Table C.1: Team Information

Sr. No.	Name of Member	Email ID	Role in the Project
1	Ms. Nikita Aher	nikitaaher947@gmail.com	Backend Development for Typing Model
2	Ms. Soniya Pawar	soniya000pawar@gmail.com	Frontend (UI/UX) Development
3	Ms. Pranjali Rasal	pranjalarasal782004@gmail.com	Backend Development for Voice Model
4	Ms. Sakshi Wagh	sakshiwagh527@gmail.com	Backend Development for Facial Recognition Model

Bibliography

- [1] Kovačević, Nikola, et al. "Personality trait recognition based on smartphone typing characteristics in the wild." *IEEE Transactions on Affective Computing* 14.4 (2023): 3207-3217.
- [2] Xu, J., et al. "Prediction of the big five personality traits using static facial images of college students with different academic backgrounds. *IEEE Access: Practical Innovations, Open Solutions*, 9, 76822-76832." 2021,
- [3] Brahnam, Sheryl, and Loris Nanni. "Predicting trait impressions of faces using local face recognition techniques." *Expert Systems with Applications* 37.7 (2010): 5086-5093.
- [4] Ilmini, Kalani, and T. Fernando. "Persons' personality traits recognition using machine learning algorithms and image processing techniques." *Advances in Computer Science: an International Journal* 5.1 (2016): 40-44.
- [5] Kachur, Alexander, et al. "Assessing the Big Five personality traits using real-life static facial images." *Scientific Reports* 10.1 (2020): 8487.
- [6] Bhin, Hyeonuk, and Jongsuk Choi. "Multimodal Personality Recognition Using Self-Attention-Based Fusion of Audio, Visual, and Text Features." *Electronics* 14.14 (2025): 2837.
- [7] Ahmad H, Asghar MU, Asghar MZ, Khan A, Mosavi AH (2021) A hybrid deep learning technique for personality trait classification from text. *IEEE Access* 9:146214–146232.
- [8] Ouarka, Ayoub, et al. "A deep multimodal fusion method for personality traits prediction." *Multimedia Tools and Applications* (2024): 1-23.
- [9] Alam, Firoj, and Giuseppe Riccardi. "Predicting personality traits using multimodal information." *Proceedings of the 2014 ACM multi media on workshop on computational personality recognition*. 2014.
- [10] Yesu, Kolhandai, et al. "Big five personality traits inference from five facial shapes using CNN." 2021 *IEEE 4th International Conference on Computing, Power and Communication Technologies (GUCON)*. IEEE, 2021.